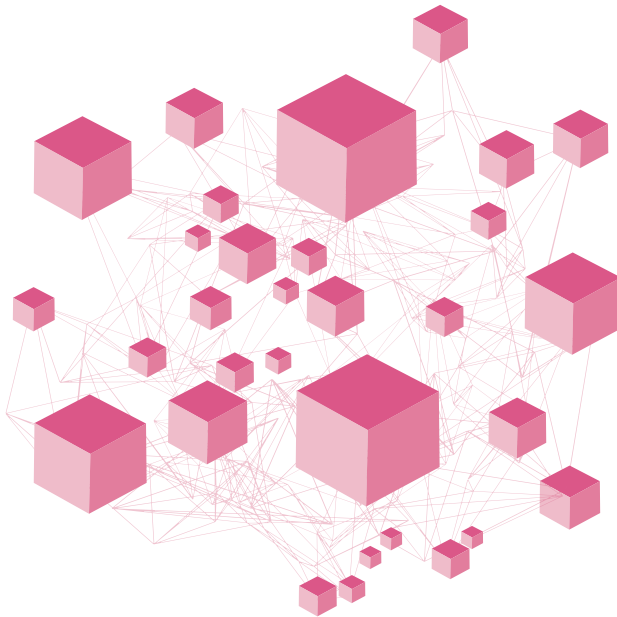




Smart Contracts

Programming in a Blockchain:

An Overview



Blockchain technology is being adopted worldwide for secure and decentralized applications (dApps) in domains like e-governance, cryptocurrencies, smart contracts, logistics, social media, and so on. Let us have a look at the platforms that can be used to build a smart contract.

Blockchains use distributed ledger technology (DLT) for secure and private transactions. These distributed ledgers in the blockchain environment prevent third-party manipulation by replicating digital assets over numerous nodes and keeping them in sync in real-time. Distributed ledger technology makes use of smart

contracts programming, using which secured apps can be developed.

Using blockchain technology's distributed ledger, a bank, for instance, can provide a higher level of security to its customers with the integration of decentralised storage. If the bank has millions of customers, then all the financial data related to those clients will be stored on millions of different computers and gadgets to enforce security and decentralisation. In contrast, when data is stored in one centralised server or single data centre, then a hacker can penetrate into the server and steal sensitive client information. That's why it's so important for government entities to decentralise their online apps. Land registrations, sensitive citizen information (including Aadhaar and PAN cards), etc, can all be safeguarded on government servers that utilise decentralized apps built on blockchain technology.

By using cryptography and distributed ledger technology, blockchain technology makes it extremely difficult for any outside party to falsify, alter, or otherwise manipulate recorded data. Each machine in the blockchain network has a copy of the digital ledger of transactions. There are several transactions in each block in the chain. Every time a new transaction takes place on the blockchain, it is recorded in the ledgers of all participants on the chain. In DLT, a cryptographic hash is used to record transactions in an immutable fashion so that multi-level security can be provided to them. This makes it difficult for hackers to alter the whole chain with only a single block alteration, since they would have to do so for every existing version of the chain. As additional blocks are added to a blockchain, like those used by Bitcoin and Ethereum, the ledger becomes increasingly secure.

A few use cases and applications of DLT and dApps in blockchain are:

- Non-fungible tokens (NFTs)
- Cryptocurrencies
- International financial transactions
- Gaming applications
- Artificial intelligence
- Cybersecurity
- E-commerce platforms
- Data privacy
- Secured personal information
- Smart contracts
- Decentralized finance (DeFi)
- E-voting in elections
- Media services
- Internet of Things (IoT)
- E-governance
- Digital notary
- Multimedia data protection
- Intellectual properties and digital rights